

Apex Diagnostics

Comprehensive Pre-Marital Health Report

Clinical Profile: Male Teratozoospermia

Name: Mr. Vikram Singh

Age/Sex: 32 Yrs / M

Date: 02 Jun 2026, 03:50 PM

ID: ADX-27150

BLOOD GROUP - ABO TYPING

Parameter	Result	Reference	Unit
Blood Group (A / B / AB / O)	O	-	-

BLOOD GROUP - RH (RHESUS) TYPING

Parameter	Result	Reference	Unit
Rh Factor (Positive / Negative)	Positive	-	-

COMPLETE BLOOD COUNT (CBC) / HEMOGRAM

Parameter	Result	Reference	Unit
Total Red Blood Cell Count (RBC)	4.7	4.5-5.9	mill/μL
Hemoglobin (Hb)	15.9	13.5-17.5	g/dL
Hematocrit / Packed Cell Volume (PCV)	40.9	41.0-53.0	%
Mean Corpuscular Volume (MCV)	87.0	80.0-100.0	fL
Mean Corpuscular Hemoglobin (MCH)	30.2	27.0-33.0	pg
Mean Corpuscular Hemoglobin Concentration (MCHC)	38.9	32.0-36.0	g/dL
Red Cell Distribution Width - CV (RDW-CV)	12.9	11.5-14.5	%
Red Cell Distribution Width - SD (RDW-SD)	40.4	39.0-46.0	fL
Total White Blood Cell Count (WBC / TLC)	8.3	4.5-11.0	thou/μL
Neutrophils (%)	56.1	40-80	%
Neutrophils - Absolute Count	4.66	2.0-7.0	thou/μL
Lymphocytes (%)	27.5	20-40	%
Lymphocytes - Absolute Count	2.49	1.0-3.0	thou/μL
Monocytes (%)	4.7	2-10	%
Monocytes - Absolute Count	0.5	0.2-1.0	thou/μL
Eosinophils (%)	3.7	1-6	%
Eosinophils - Absolute Count	0.31	0.02-0.5	thou/μL
Basophils (%)	0.5	0-2	%
Basophils - Absolute Count	0.04	0.0-0.1	thou/μL
Immature Granulocytes (%)	0.2	0-0.5	%
Immature Granulocytes - Absolute Count	0.01	0-0.03	thou/μL
Nucleated Red Blood Cells (NRBC) - Absolute Count	0.0	0.0	thou/μL
Nucleated Red Blood Cells (NRBC) %	0.0	0.0	%
Platelet Count	245	150-450	thou/μL
Mean Platelet Volume (MPV)	9.5	7.5-11.5	fL
Platelet Distribution Width (PDW)	12.0	9.0-17.0	fL
Plateletcrit (PCT)	0.25	0.15-0.35	%
Platelet-Large Cell Ratio (P-LCR)	25.0	15.0-35.0	%

ERYTHROCYTE SEDIMENTATION RATE (ESR)

Parameter	Result	Reference	Unit
Erythrocyte Sedimentation Rate (ESR)	10	< 15	mm/hr

BLOOD SUGAR - FASTING (FBS / FBG)

Parameter	Result	Reference	Unit
Fasting Blood Glucose (FBG)	94.3	70-100	mg/dL

BLOOD SUGAR - RANDOM (RBS / BSR)

Parameter	Result	Reference	Unit
Random Blood Sugar (RBS)	105.6	70-140	mg/dL

GLYCATED HEMOGLOBIN (HBA1C)

Parameter	Result	Reference	Unit
HbA1c (%)	5.3	4.0-5.6	%
Estimated Average Glucose (eAG)	105.4	70-126	mg/dL

AVERAGE BLOOD GLUCOSE (ABG)

Parameter	Result	Reference	Unit
Average Blood Glucose (ABG)	109.3	< 117	mg/dL

LIVER FUNCTION TEST (LFT) - TOTAL BILIRUBIN

Parameter	Result	Reference	Unit
Total Bilirubin	0.81	0.3-1.2	mg/dL
Direct (Conjugated) Bilirubin	0.2	< 0.3	mg/dL
Indirect (Unconjugated) Bilirubin	0.2	< 0.3	mg/dL

LIVER FUNCTION TEST (LFT) - SGPT / ALT

Parameter	Result	Reference	Unit
Alanine Aminotransferase (SGPT / ALT)	23	< 41	U/L

LIVER FUNCTION TEST (LFT) - SGOT / AST

Parameter	Result	Reference	Unit
Aspartate Aminotransferase (SGOT / AST)	28	< 40	U/L

KIDNEY FUNCTION TEST (KFT)

Parameter	Result	Reference	Unit
Serum Creatinine	0.87	0.6-1.2	mg/dL
Blood Urea Nitrogen (BUN)	12	7-20	mg/dL
BUN / Creatinine Ratio	13.8	10-20	Ratio
Serum Uric Acid	5.4	3.5-7.2	mg/dL
Serum Urea	26	15-40	mg/dL
Estimated Glomerular Filtration Rate (eGFR)	112	> 90	mL/min
Serum Sodium (Na ⁺)	140	135-145	mEq/L
Serum Potassium (K ⁺)	4.2	3.5-5.1	mEq/L
Serum Chloride (Cl ⁻)	102	98-107	mEq/L
Serum Bicarbonate (HCO ₃ ⁻)	26	22-29	mEq/L

SERUM IRON

Parameter	Result	Reference	Unit
Serum Iron	95	50-170	μg/dL

TOTAL IRON BINDING CAPACITY (TIBC)

Parameter	Result	Reference	Unit
Total Iron Binding Capacity (TIBC)	306	250-450	μg/dL

UNSATURATED IRON BINDING CAPACITY (UIBC)

Parameter	Result	Reference	Unit
Unsaturated Iron Binding Capacity (UIBC)	211	110-370	μg/dL

TRANSFERRIN SATURATION (%)

Parameter	Result	Reference	Unit
Transferrin Saturation (%)	31	20-50	%

SERUM FERRITIN

Parameter	Result	Reference	Unit
Serum Ferritin	73	20-250	ng/mL

HEMOGLOBIN HPLC / VARIANT ANALYSIS			
Parameter	Result	Reference	Unit
Hemoglobin A (HbA)	96.8	> 95.0	%
Hemoglobin A2 (HbA2)	2.5	1.5-3.5	%
Hemoglobin F (HbF / Fetal Hemoglobin)	0.5	< 2.0	%
Hemoglobin S (HbS)	0.0	0.0	%
Hemoglobin C (HbC)	0.0	0.0	%
Hemoglobin D (HbD)	0.0	0.0	%
Hemoglobin E (HbE)	0.0	0.0	%
Total Hemoglobin (g/dL)	15.9	12.0-16.0	g/dL
Unknown / Other Hemoglobin Fractions	0.0	0.0	%
Interpretation / Diagnosis	Normal Study	Normal Study	-

HEMOGLOBIN A2			
Parameter	Result	Reference	Unit
Hemoglobin A2 (HbA2) %	2.5	1.5-3.5	%

HEMOGLOBIN C			
Parameter	Result	Reference	Unit
Hemoglobin C (HbC) %	0.0	0.0	%

HEMOGLOBIN D			
Parameter	Result	Reference	Unit
Hemoglobin D (HbD) %	0.0	0.0	%

HEMOGLOBIN F			
Parameter	Result	Reference	Unit
Hemoglobin F / Fetal Hemoglobin (HbF) %	0.5	< 2.0	%

HEMOGLOBIN S			
Parameter	Result	Reference	Unit
Hemoglobin S (HbS) %	0.0	0.0	%

THYROID STIMULATING HORMONE (TSH)			
Parameter	Result	Reference	Unit
Thyroid Stimulating Hormone (TSH / uTSH)	1.33	0.4-4.0	µIU/mL
Triiodothyronine (T3) - Total	120	80-200	ng/dL
Thyroxine (T4) - Total	8.0	5.0-12.0	µg/dL
Free T3 (FT3)	3.5	2.3-4.2	pg/mL
Free T4 (FT4)	1.2	0.9-1.7	ng/dL

VITAMIN B12 (CYANOCOBALAMIN)			
Parameter	Result	Reference	Unit
Vitamin B12 (Cobalamin) - Serum	573	200-900	pg/mL

LIPID PROFILE / LIPID SCREEN			
Parameter	Result	Reference	Unit
Total Cholesterol	177	< 200	mg/dL
High-Density Lipoprotein Cholesterol (HDL-C)	46	> 40	mg/dL
Low-Density Lipoprotein Cholesterol (LDL-C)	84	< 100	mg/dL
Very Low-Density Lipoprotein (VLDL-C)	15	2-30	mg/dL
Triglycerides	78	< 150	mg/dL
Non-HDL Cholesterol	131	< 130	mg/dL
Total Cholesterol / HDL Ratio (TC:HDL)	3.8	< 5.0	Ratio
LDL / HDL Ratio	1.8	< 3.0	Ratio

SYPHILIS - VDRL / RPR			
Parameter	Result	Reference	Unit
VDRL / RPR Result (Reactive / Non-Reactive)	Non-Reactive	Non-Reactive	-
RPR Titre (if reactive)	Not Applicable	Not Applicable	-

HEPATITIS B SURFACE ANTIGEN (HBSAG)			
Parameter	Result	Reference	Unit
HBsAg - Qualitative Result (Reactive / Non-Reactive)	Non-Reactive	Non-Reactive	-
HBsAg - OD Value / Signal-to-Cutoff Ratio	0.2	< 0.9	S/CO

HEPATITIS C VIRUS ANTIBODY (ANTI-HCV)			
Parameter	Result	Reference	Unit
Anti-HCV Antibody - Qualitative Result (Reactive / Non-Reactive)	Non-Reactive	Non-Reactive	-
Anti-HCV - OD Value / Signal-to-Cutoff Ratio	0.1	< 0.9	S/CO

HIV 1 & 2 ANTIBODY TEST			
Parameter	Result	Reference	Unit
HIV 1 & 2 Antibody - Result (Reactive / Non-Reactive)	Non-Reactive	Non-Reactive	-
HIV ICT (Immunochromatographic Test) Result	Non-Reactive	Non-Reactive	-

HIV COMBO (P24 ANTIGEN + ANTIBODY) TEST			
Parameter	Result	Reference	Unit
HIV P24 Antigen - Result (Detected / Not Detected)	Not Detected	Not Detected	-
HIV 1 & 2 Antibody - Result (Reactive / Non-Reactive)	Non-Reactive	Non-Reactive	-
HIV Combo Assay - OD Value / S/CO Ratio	0.1	< 0.9	S/CO
HIV 1 Differentiation Result	Negative	Negative	-
HIV 2 Differentiation Result	Negative	Negative	-

URINE ROUTINE & MICROSCOPY EXAMINATION			
Parameter	Result	Reference	Unit
Colour	Pale Yellow	Pale Yellow	-
Appearance / Clarity / Turbidity	Clear	Clear	-
Specific Gravity	1.015	1.005-1.030	-
pH	6.0	4.5-8.0	-
Urinary Protein	Absent	Absent	-
Urinary Glucose	Absent	Absent	-
Urinary Ketones	Absent	Absent	-
Urinary Bilirubin	Absent	Absent	-
Urobilinogen	Normal	Normal	-
Blood / Haemoglobin in Urine	Absent	Absent	-
Nitrites	Absent	Absent	-
Leukocyte Esterase	Absent	Absent	-
Pus Cells (WBCs) per HPF	1-2	0-5	per HPF
Red Blood Cells (RBCs) per HPF	0-1	0-2	per HPF
Epithelial Cells	Rare	Rare	per HPF
Casts	28	< 40	U/L
Crystals	Absent	Absent	-
Bacteria	Absent	Absent	-
Yeast / Fungi	28	< 40	U/L
Mucus Threads	Absent	Absent	-
Amorphous Material	6.0	4.5-8.0	-

SEMEN ANALYSIS

Parameter	Result	Reference	Unit
Semen Volume	2.6	>= 1.5	mL
Semen pH	7.5	>= 7.2	-
Colour / Appearance	Pale Yellow	Pale Yellow	-
Viscosity	Normal	Normal	-
Liquefaction Time	25	< 60	mins
Sperm Concentration	57	>= 15	mill/mL
Total Sperm Count per Ejaculate	148	>= 39	mill/ ejac.
Total Sperm Motility (%)	58	>= 42	%
Progressive Motility (PR) %	59	>= 30	%
Non-Progressive Motility (NP) %	59	>= 30	%
Immotility (IM) %	42	-	%
Total Motile Sperm Count (TMSC)	86	-	million
Sperm Morphology - Normal Forms (%)	6.0	4.5-8.0	-
Sperm Head Defects (%)	60	-	%
Sperm Midpiece Defects (%)	20	-	%
Sperm Tail Defects (%)	18	-	%
Sperm Vitality / Viability (%)	75	>= 54	%
Pus Cells / Leucocytes (per HPF)	1-2	0-5	per HPF
Round Cells per mL	< 1	< 1	mill/mL
Sperm Agglutination	Absent	Absent	-
Interpretation / Fertility Prognosis	Teratozoospermia	Normozoospermia	-